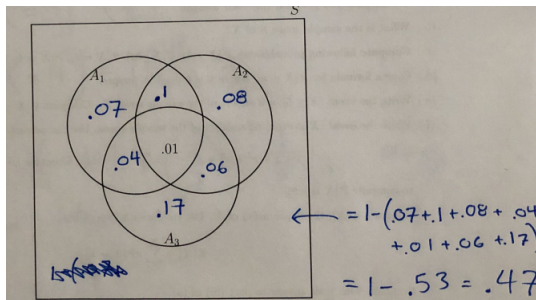


13 2026-02-13 | Week 05 | Lecture 13

We went over some example problems:

- The solution to problem 1 on homework 2. The key idea involves correctly filling out the venn diagram as shown:



- The solution to problem 14 on homework 4.
- An example about computing probabilities involving the phrase “at least” (shown below).

Example 57. Suppose we roll a 6-sided dice three times. What’s the probability that we get at least one 6?

Solution: The key idea is the following

$$\mathbb{P}[\text{at least one six is rolled}] = 1 - \mathbb{P}[\text{no sixes are rolled}]$$

To put this idea into practice, let $A_1 = [\text{the first dice roll is a 6}]$, $A_2 = [\text{the second dice roll is a 6}]$, and $A_3 = [\text{the third dice roll is a 6}]$. Clearly, A_1, A_2, A_3 are independent because the dice rolls don’t influence each other. Then

$$\begin{aligned} \mathbb{P}[\text{at least one six is rolled}] &= \mathbb{P}[A_1 \cup A_2 \cup A_3] \\ &= 1 - \mathbb{P}[A_1^c \cap A_2^c \cap A_3^c] && \text{by de Morgan's law} \\ &= 1 - \mathbb{P}[A_1^c] \mathbb{P}[A_2^c] \mathbb{P}[A_3^c] && \text{by independence} \\ &= 1 - \left(\frac{5}{6}\right) \left(\frac{5}{6}\right) \left(\frac{5}{6}\right) \\ &= 1 - \left(\frac{5}{6}\right)^3. \end{aligned}$$

End of Example 57. $\square$